

Does Age Shape How Athletes Move?

Extending the Quantile Cube Framework to Detect Age Effects in Movement Distribution Profiles

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Joint work with Dr. Jan Hannig

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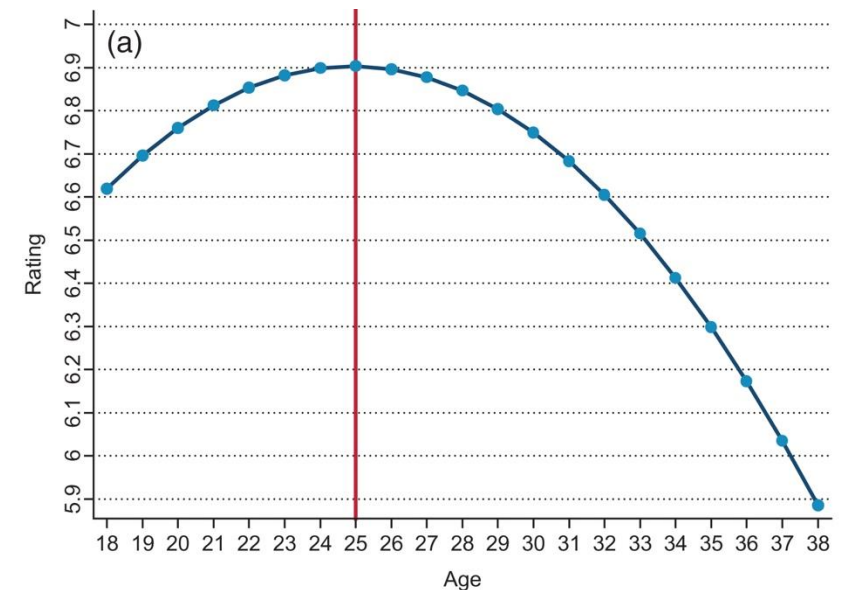


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Motivation: Beyond the Aging Curve



- Standard aging curves relate age to average performance outcomes.^[1, 2, 3]
- Recent methodological advances refine estimates of these curves, but the underlying objective stays the same.^[4, 5]
- Soccer-specific aging work (peak age 25–27) faces the same limit: two players can have identical output and completely different movement profiles.^[6]



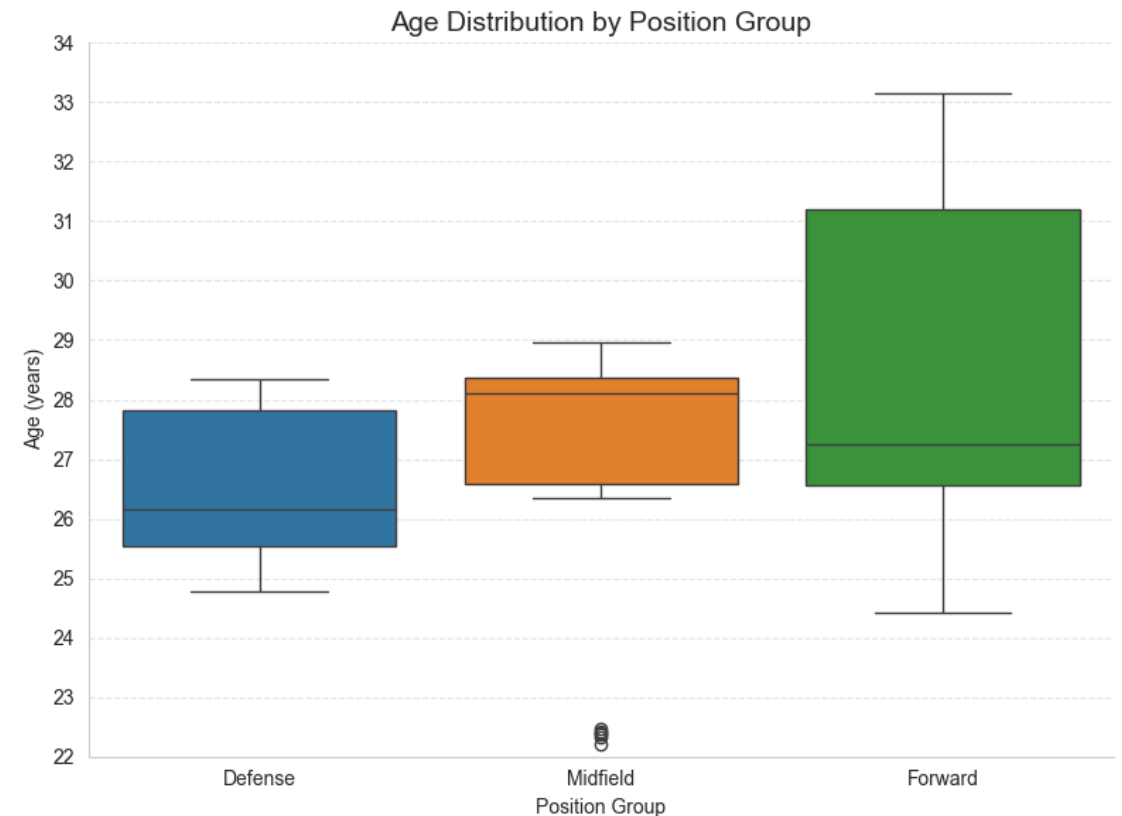
[Fig. adapted from [6] Dendir (2016), *Journal of Sports Analytics*]

Does age shape *how* an athlete moves, not just how well they perform, independent of playing time and position?

The Data



- GPS tracking data from one season for a professional women's soccer team (IRB 25-1234)
 - Sampling at 100 Hz with access at 10 Hz: timestamp, longitude, latitude
- Retained athlete-match sessions with ≥ 60 minutes played in ≥ 3 matches
 - Total: 17 athletes, 28 matches $\rightarrow$ 216 athlete-match sessions
- Covariates:
 - standardized age (mean 27.4 yrs, SD 2.8; range 22.2 – 33.1)
 - position (6 defenders, 4 midfielders, 7 forwards)

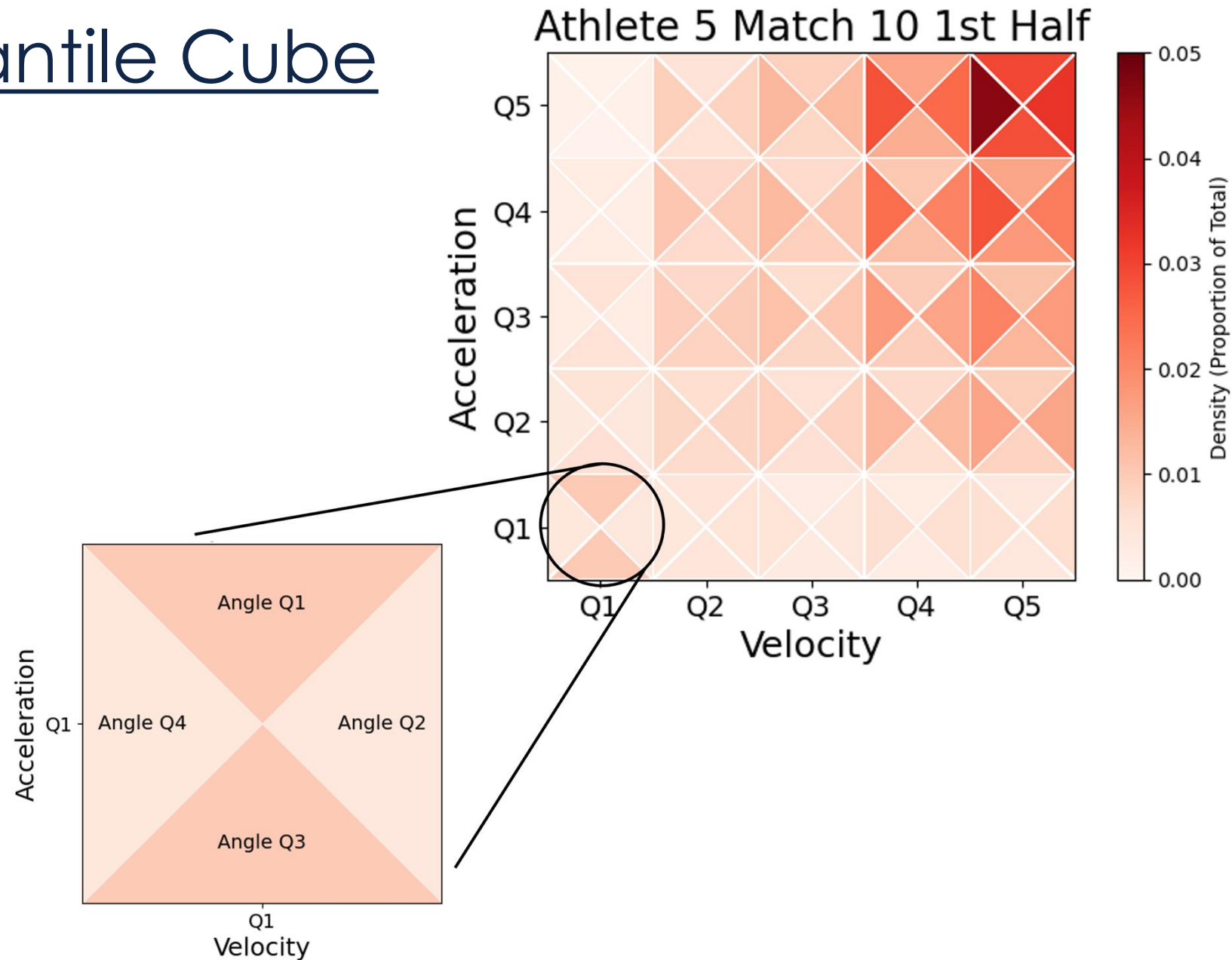


The Quantile Cube



- Following Thomas and Hannig (2026)^[7], we constructed a quantile cube, a 3-D summary representation of movement distributions, for each athlete-match session
- **Construction**
 1. **Derived quantile cutoffs using the full dataset**
 - 5 quantiles for Velocity and Acceleration
 - 4 quantiles for angle, starting from a shifted -41°
 2. **Constructed a quantile cube for each athlete-match**
 - 100-dimensional vector representing deciseconds of time spent in a certain movement category
 3. **Result: $Y_{n \times d}$**
 - $n = 216$ rows: quantile cube for an athlete-match
 - $d = 100$ features: time spent within the corresponding movement profile

The Quantile Cube



The Model



- Within a single quantile cube, bin counts span orders of magnitude.
- A standard approach does not fit the whole vector: NB is exact but costly at large counts; Gaussian is fast but breaks down near zero
- Our Approach: partition each bin j by its expected count relative to a threshold τ

$$y_{ij} \sim \begin{cases} \text{NB}\left(\alpha_{ij}t_i, \frac{v_j}{1+v_j}\right) & \text{if } \mu_{ij}(\boldsymbol{\theta}_0) < \tau, \\ \mathcal{N}\left(\mu_{ij} = \frac{\alpha_{ij}t_i}{v_j}, \sigma_{ij}^2 = \frac{\alpha_{ij}t_i(v_j+1)}{v_j^2}\right) & \text{if } \mu_{ij}(\boldsymbol{\theta}_0) \geq \tau \end{cases}$$

where $\alpha_{ij} = \exp(X_i^T \beta_j)$ and the parameter vector $\boldsymbol{\theta} = (\beta_1^T, \dots, \beta_{100}^T, v_1, \dots, v_{100})^T$

The Generalized Fiducial Density



- GFI expresses the data through a structural equation $Y = G(U, \theta)$ involving the parameters θ and auxiliary variables $\mathbf{U}$ with known distribution^[8, 9]

- **Theorem 1:** The Joint Generalized Fiducial Density takes the form

$$r(\theta | \mathbf{y}) \propto L(\mathbf{y} | \theta) J(\theta, \mathbf{y}_C)$$

- **Proposition 2 (Block-Diagonal Jacobian):** The Jacobian decomposes into one block per movement bin, where each block depends only on that bin's own parameters $\theta_j = (\beta_j^T, v_j)$.

$$J_C(\theta) = \partial Y_C / \partial \theta = \text{diag}(J_1, J_2, \dots, J_{100})$$

$$J(\theta, \mathbf{y}_C) = \sqrt{\det(n^{-1} J_C(\theta)^T J_C(\theta))} = \prod_{j=1}^{100} \sqrt{\det(n^{-1} J_j^T J_j)}$$

Fitting the Model



- **Parameter vector:**

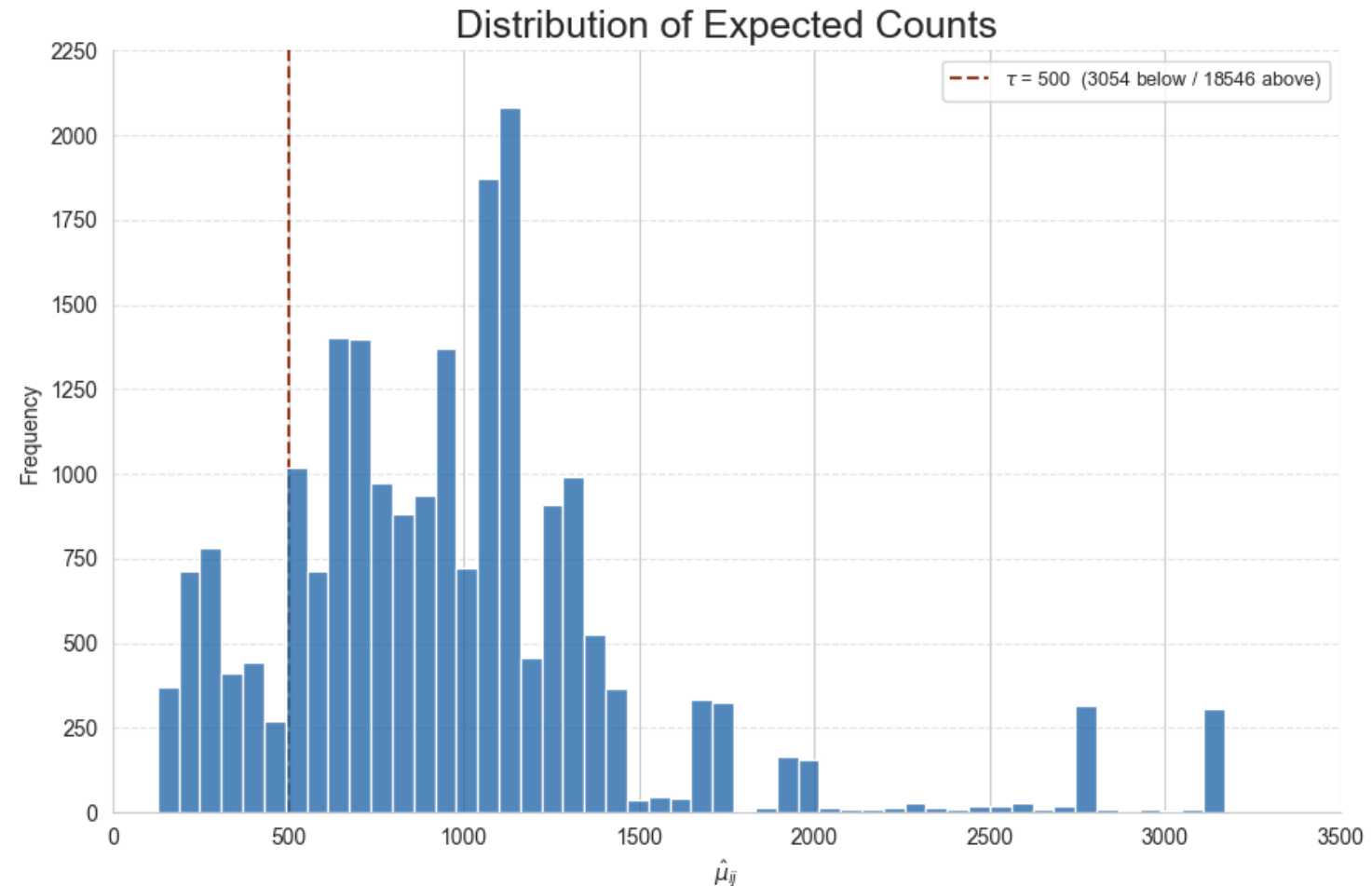
$$\theta = (\beta_1^\top, \dots, \beta_{100}^\top, v_1, \dots, v_{100})^\top$$

→ each of the 100 movement bins has its own regression coefficients and dispersion parameter

- **Partition Rule:** route bin j to the Normal component when the estimated expected count

$$\hat{\mu}_{ij} \geq \tau = 500$$

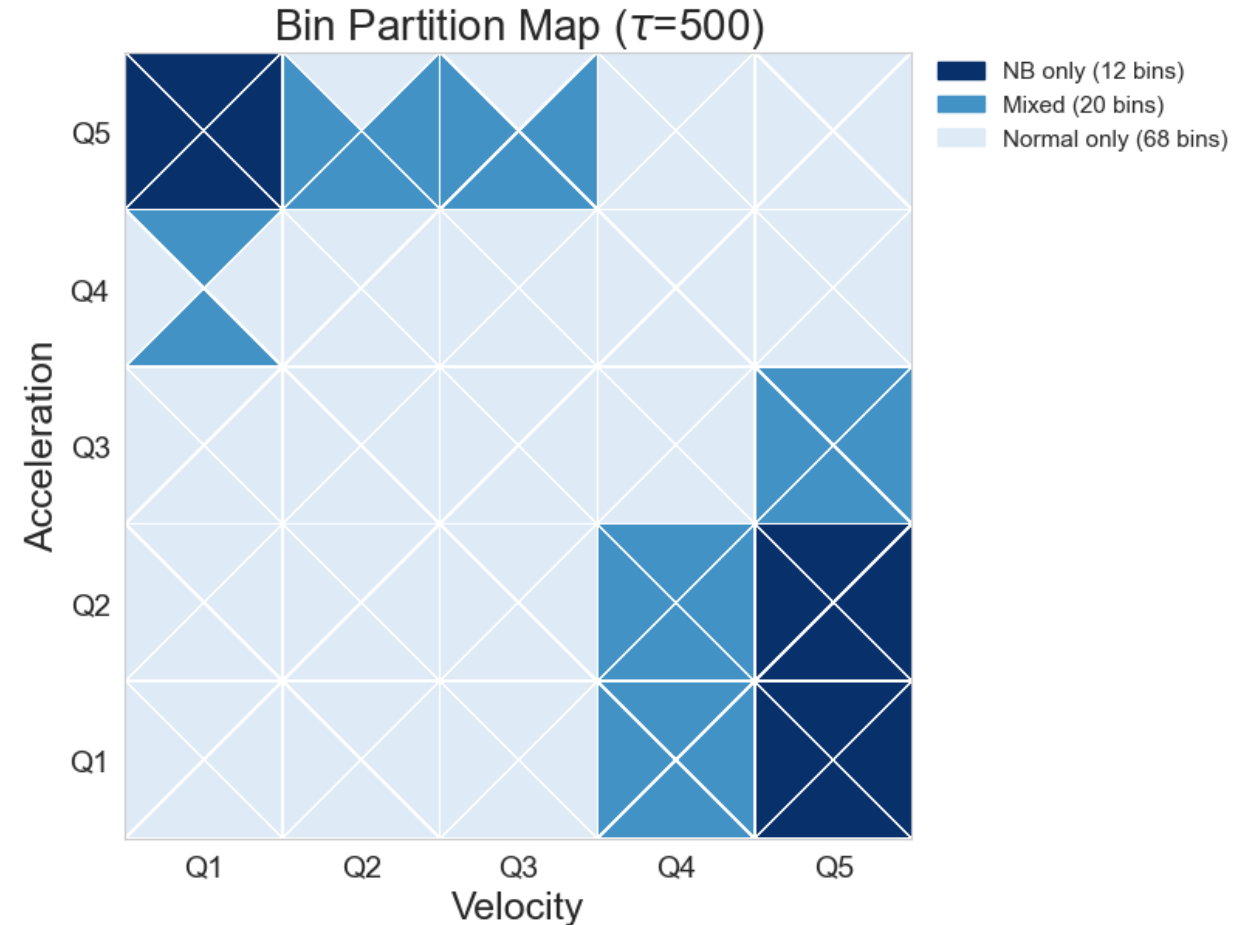
- Estimated via pooled rate scaled by exposure: $\hat{\mu}_{ij} = \hat{r}_j t_i$
with $\hat{r}_j = n^{-1} \sum_i Y_{ij} / t_i$



Fitting the Model



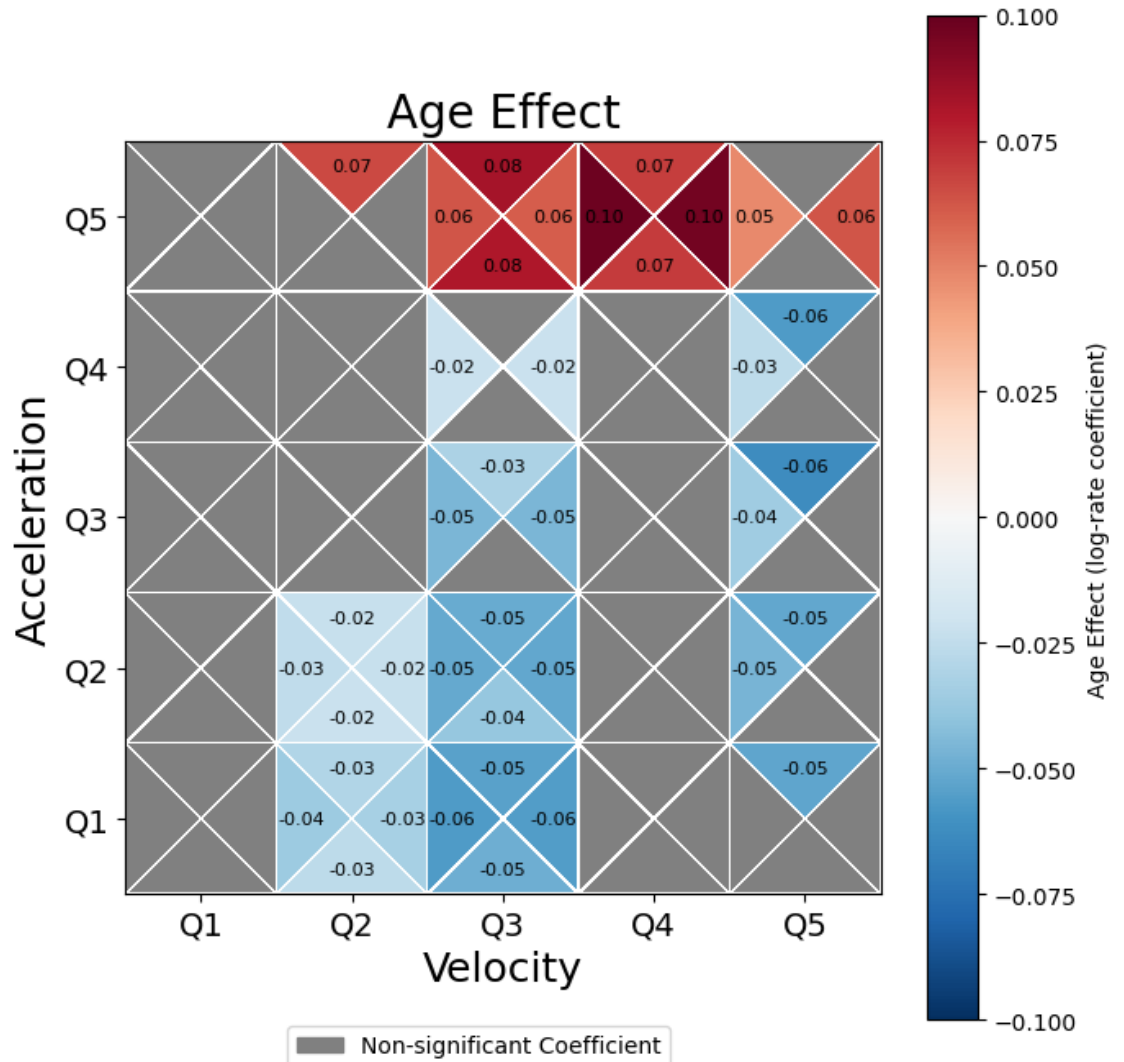
- **Resulting Partition:** 12 bins entirely Negative Binomial, 68 entirely Normal, 20 mixed across sessions
- **Algorithm:** Block-Diagonal Jacobian allows each bin to be fit independently via Hamiltonian Monte Carlo
 - 4 chains, 1000 warmup / 2000 sampling
 - All 100 bins reached $\hat{R} < 1.01$
- **Multiple Testing:** 100 bins $\times$ each covariate = 100 tests per covariate $\rightarrow$ Benjamini–Hochberg correction^[10] at FDR 0.05



Results: The Age Effect



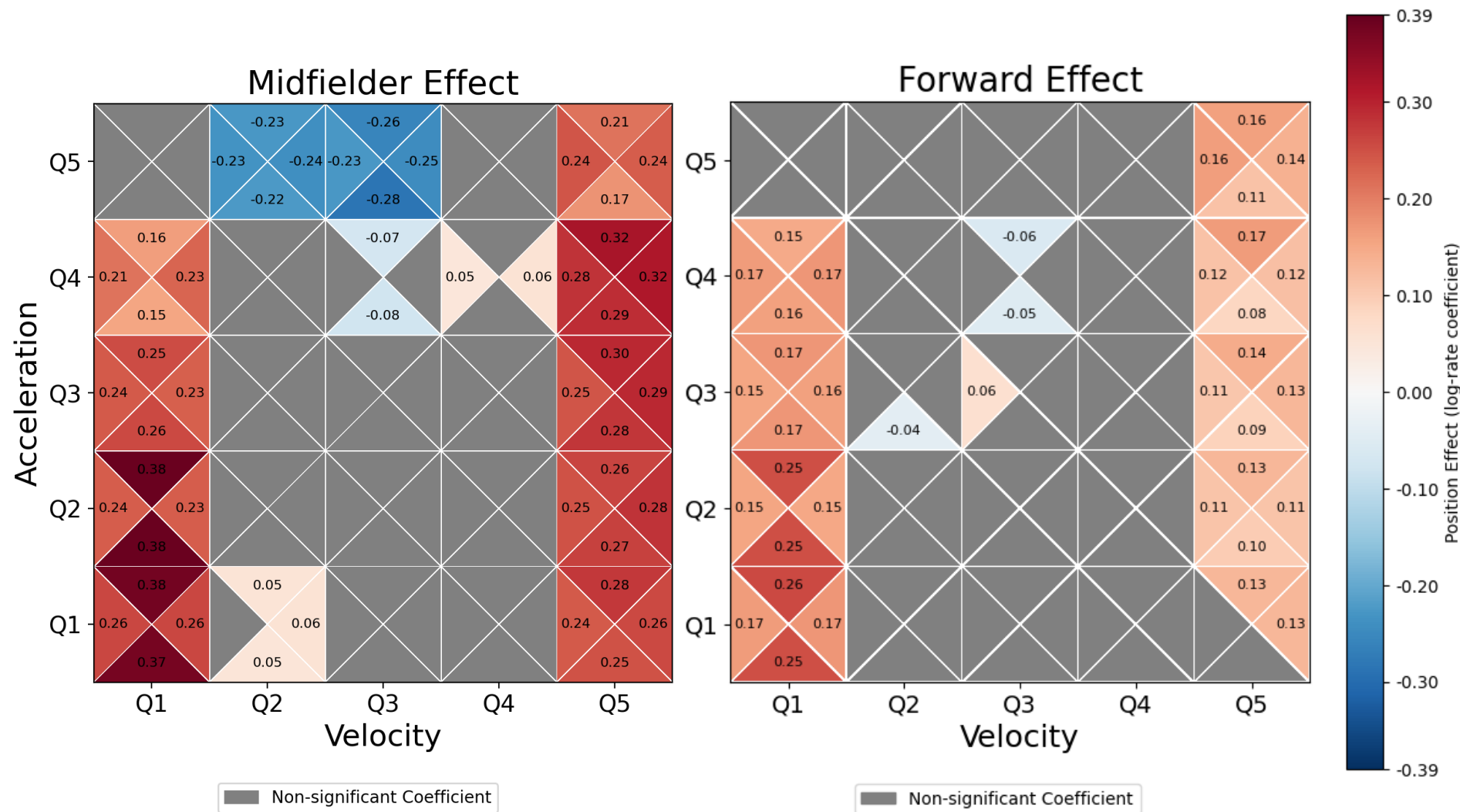
- Age effect significant (BH-corrected, FDR 0.05) in 50 of 100 movement bins
- Older athletes accumulate
 - more time in peak acceleration bins across mid-to-high velocity bins
 - less time in mid-and-high velocity across low-to-mid-acceleration bins
- Magnitude: $|\beta_{jk}| \leq 0.10$ - at most an 11% shift in expected time-density per movement bin



Results: Position Effect



- Position effects are **substantially larger**
- Position significance: forward vs. defender significant in 37 bins; midfielder vs. defender significant in 51 bins (both BH-corrected, FDR 0.05)



Conclusion



Age shapes how athletes move, not just how they perform.

Hybrid GFI model for
high-dimensional,
mixed discrete-
continuous count
data



Applied to
professional women's
soccer: "**the veteran-
efficiency effect**"

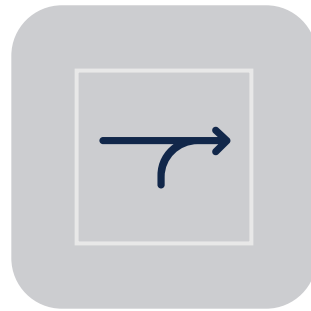


Reframes aging-curve
research beyond
outcomes to the
movement that
produces them

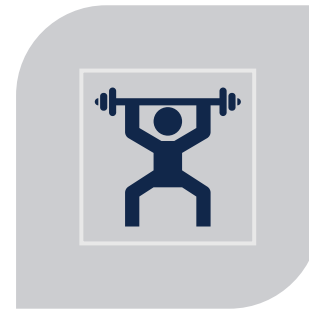
Ongoing and Future Work



**SMOOTH BASIS-
FUNCTION
EXTENSION**



**BERNSTEIN-VON
MISES THEOREM^[11]**



**WITHIN-ATHLETE
RANDOM EFFECTS**



**EXTENDING THE
FRAMEWORK**

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Thank you! Questions?

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Quantile Cube
Paper

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